



Cloud Computing

Dr. Ramkumar Krishnamoorthy

Contents

01

Vulnerabilities and Attacks

- ✓ Hypervisor Security, Storage, Hardware, and Network, Cloud Security Mechanisms

02

Data Security

- ✓ Digital Signature, Hashing, Virtualization Security

03

Privacy and Security in Cloud Storage Services

- ✓ Cloud Data Protection Models
- ✓ Security and Privacy Properties in Multi clouds.

04

Summary

- ✓ Revision



Cloud Terms

- **Cloud Client**

It includes Web browsers, Mobile Apps, Thin Clients, Terminal etc.

- **Cloud Computing Provider**

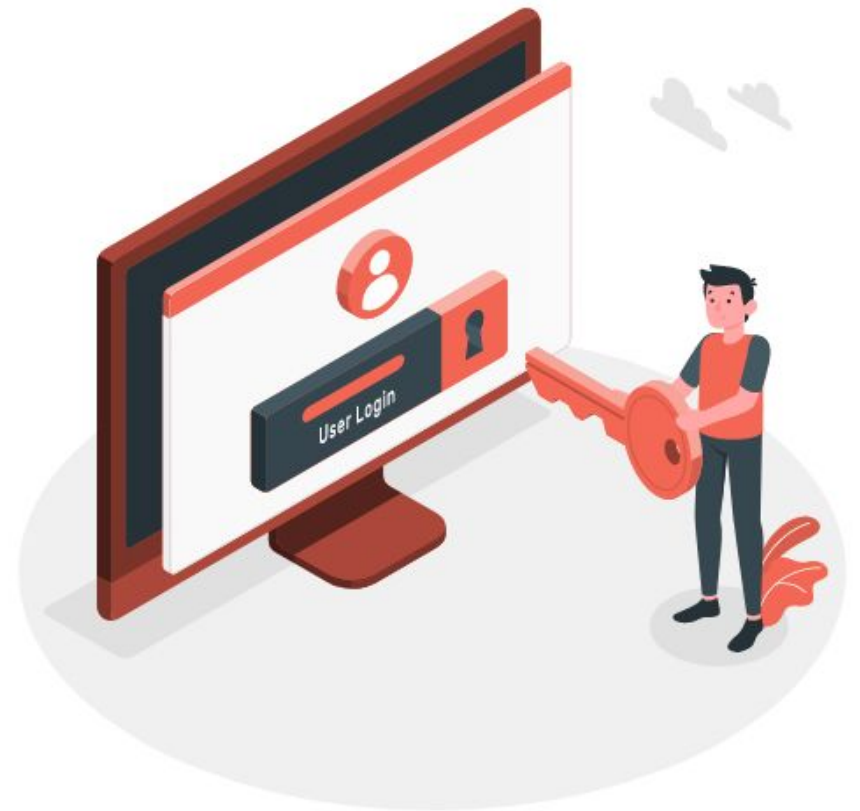


- **Cloud Platform**

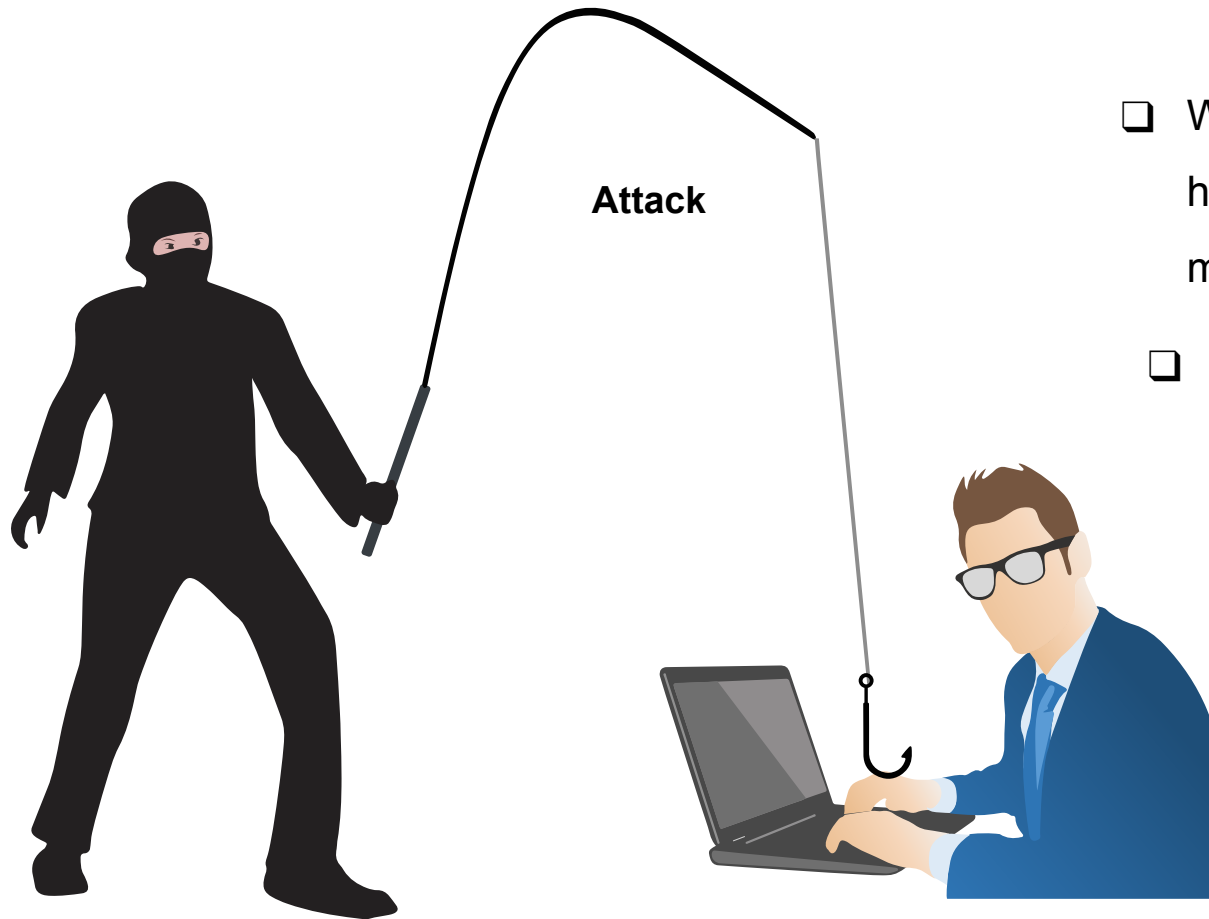


Security

- Protection of a person, building, organization, or country against threats such as crime or attacks

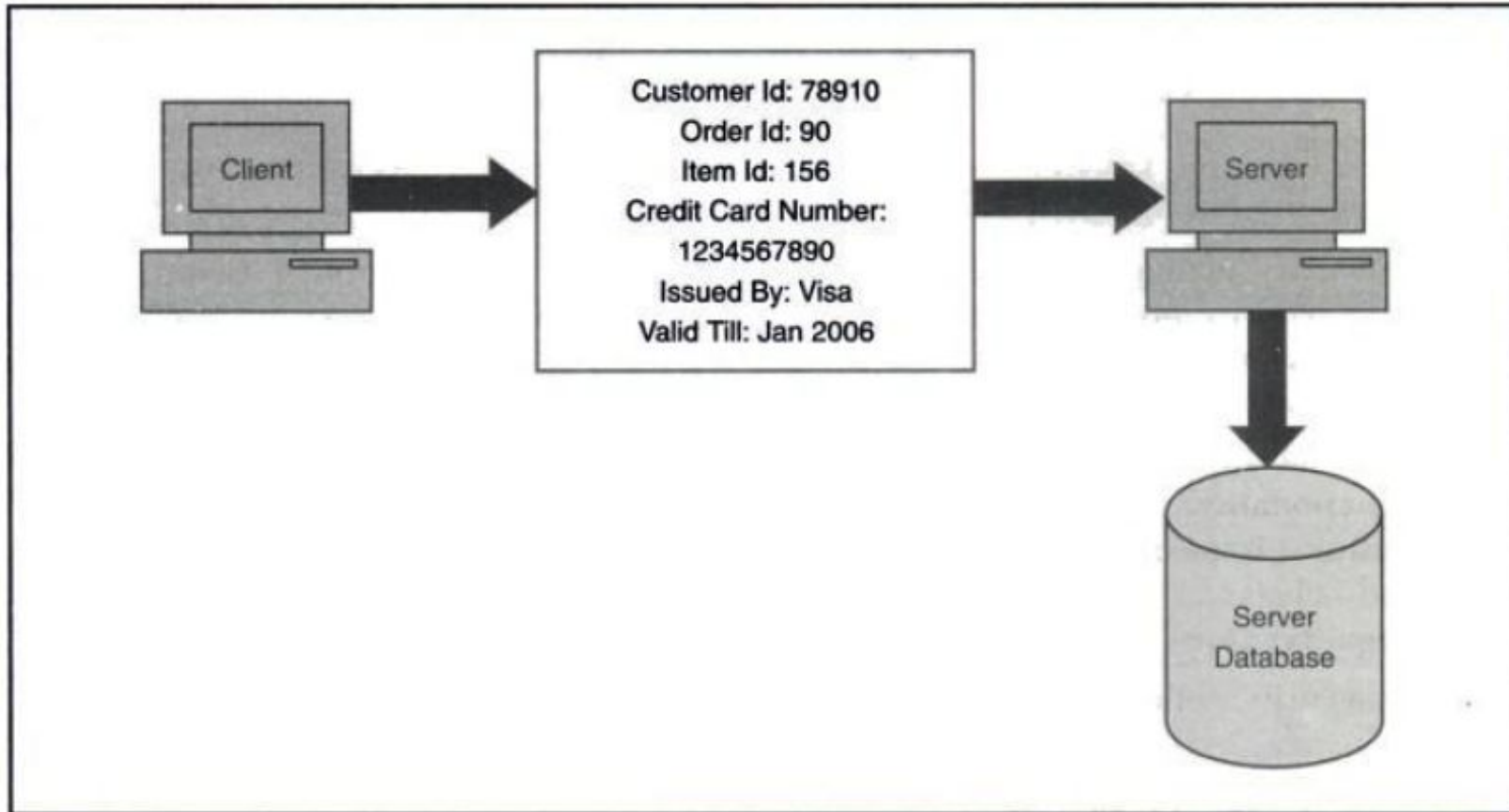


Need of Security



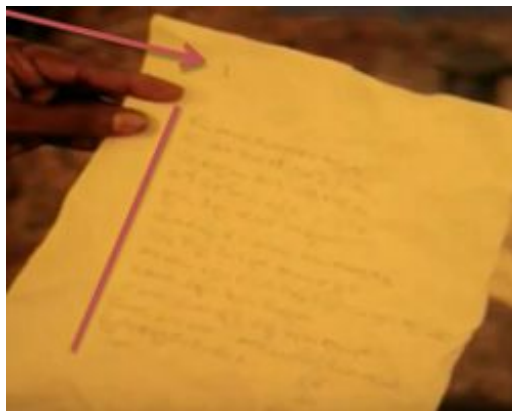
- ❑ When Computer Applications were designed to handle Financial and Personal data, Security is must
- ❑ People realized that Computerized data are most important

Example for Security holes



Example of information traveling from a client to a server over the Internet

Real-Time Example



Principles of Security

The following are the main principles of security. They are

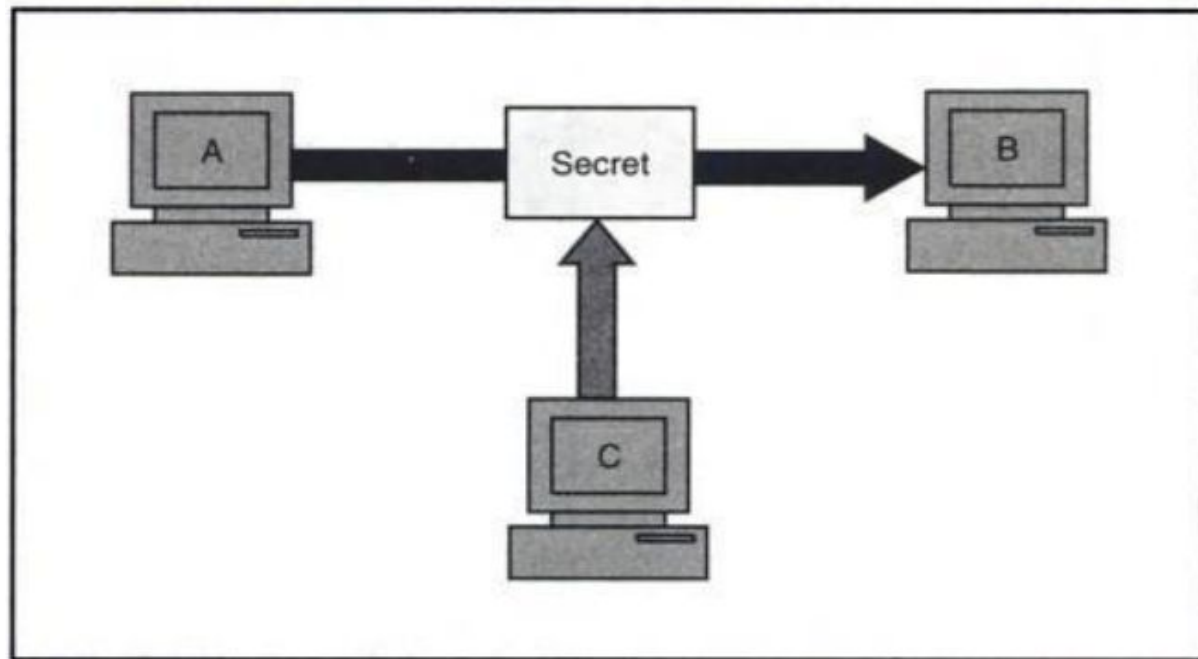
1. Confidentiality
2. Authentication
3. Integrity
4. Non-repudiation
5. Access Control
6. Availability

Principles of Security – Confidentiality (1)

- It specifies that only the sender and receiver should be able to access the message
- Example: sending an email
- **Note:** Here, C observes the email without knowing to A and B
- A failure to maintain confidentiality means that someone who shouldn't have access has managed to get it, through intentional behavior or by accident. Such a failure of confidentiality, commonly known as a *breach*, typically cannot be remedied. Once the secret has been revealed, there's no way to un-reveal it. If user bank records are posted on a public website, everyone can know user bank account number, balance, etc., and that information can't be erased from their minds, papers, computers, and other places.

Principles of Security – Confidentiality (1)

Loss of Confidentiality – **Interception**



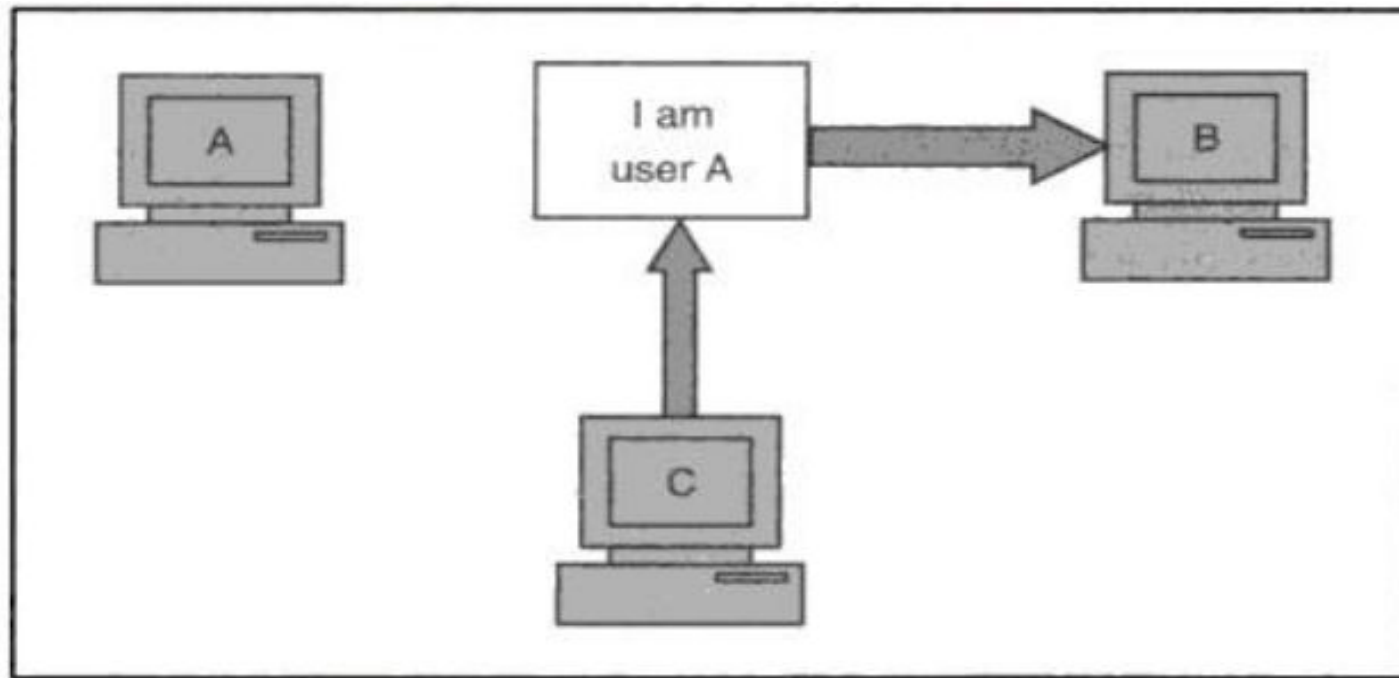
Loss of confidentiality

Principles of Security – Authentication (2)

- Authentication is used to establish a proof of identities
- Authentication is about validating user credentials like User Name/User ID and password to verify user identity.
- It ensures that the origin of the electronic message or document is correctly identified
- For instance, suppose that user C sends an electronic document over the Internet to user B
- However, the trouble is that the user C had posed as user A when she sent this document to user B
- How the B will identify, C is posing like A. that is, improper authentication will create such issue.

Principles of Security – Authentication (2)

Absence of Authentication - **Fabrication**

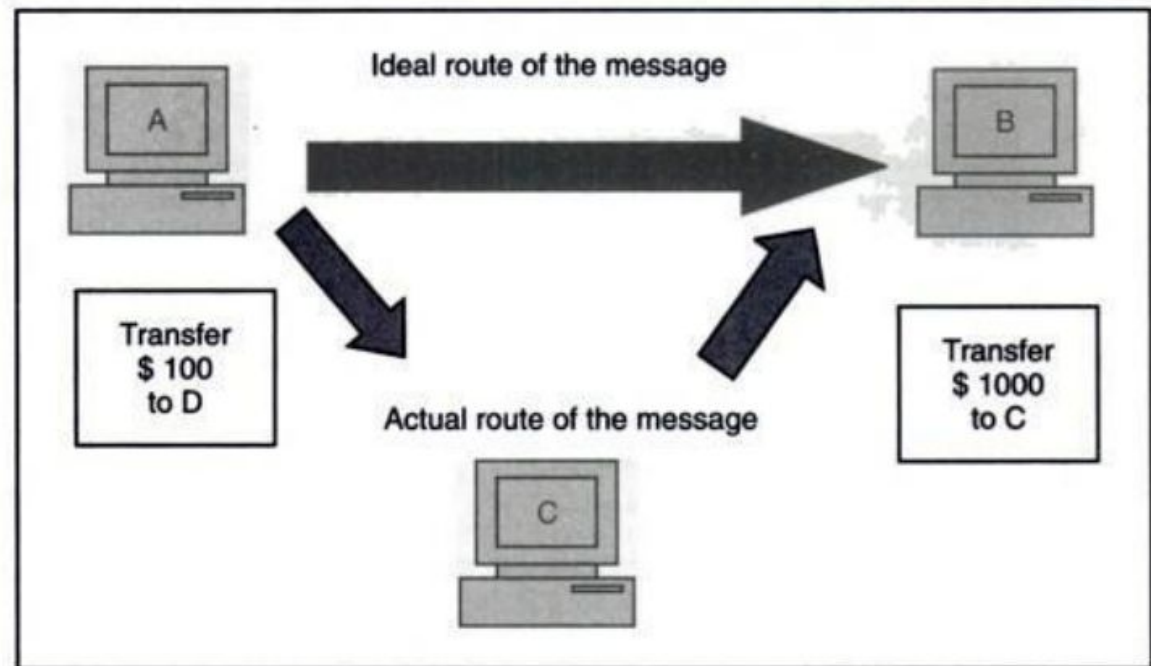


Absence of authentication

Principles of Security – Integrity (3)

- When the Contents of the Message is not changed after sender sends it and before reaches to the destination
- Example: transferring money

- Loss of Integrity – **Modification**



Loss of integrity

Principles of Security – Access Control List (4)

It is called as Access Control List (ACL). It contains the information that, who can access and who cannot. Resource access could be sharing college wifi with the students etc

- Allowing access
- Denying access
- Limiting access
- Revoking access



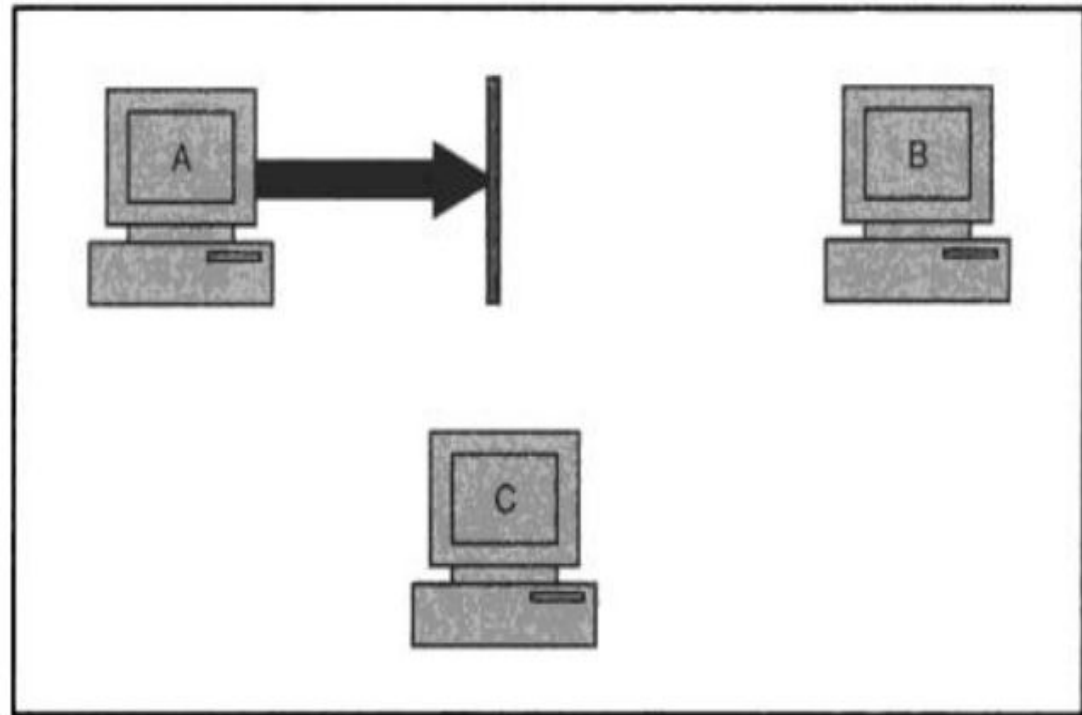
Principles of Security – Availability (5)

- The needed resources should be available for all the authorized parties whenever they send request
- The principle of availability states that resources (information) should be available to authorized parties at all times
- For example, due to the intentional actions of an unauthorized user C, an authorized user A may not be able to contact a server computer B.
- It causes the resource becomes unavailable
- Example: Granting access



Principles of Security – Availability (5)

- Attack on Availability - **Interruption**



Attack on availability

Principles of Security – Authorization (6)

- Authorization, on the other hand, occurs after user identity is successfully authenticated by the system, which ultimately gives you full permission to access the resources such as information, files, databases, funds, locations, almost anything.

Authorization - Examples

- Verifying user's permissions to access resources.
- OAuth (Open Authorization).
- Permissions –
 - Read-write access to files.
 - Allowing to access of the database.
- Specifying user roles to access data.

Authentication vs. Authorization



Authentication

Who you are



Authorization

What you can do

Some Exercises

Classify each of the following as an attack on confidentiality, integrity, and/or availability (more than one may apply). Justify user answers.

1. John copies Mary's homework
2. Paul crashes Linda's system
3. Carol changes the amount of Angelo's check from \$100 to \$1,000
4. Gina forges Roger's signature on a deed
5. Rhonda registers the domain name "AddisonWesley.com" and refuses to let the publishing house buy or use that domain name
6. Jonah obtains Peter's credit card number and has the credit card company cancel the card and replace it with another card bearing a different account number
7. Henry spoofs Julie's IP address to gain access to her computer

Vulnerability and Attacks

- In cloud computing, vulnerabilities are security weaknesses that hackers can exploit to gain unauthorized access to data and services.
- Attacks are threats that can take advantage of these vulnerabilities to cause severe consequences.
- A cloud attack is a cyber attack that targets cloud-based service platforms, such as computing services, storage services, or hosted applications in a platform as a service (PaaS) or software as a service (SaaS) model.

Vulnerability and Attacks (1)

Misconfigurations

- Meaning **flaws in the way infrastructure, services, or applications are configured**.
- Attackers can take advantage of misconfigurations by breaking into user accounts with excess permissions and using them to steal data

Denial of Service (DoS)

- It halts the server and **unable to provide services to the genuine users** also

Vulnerability and Attacks (2)

Poor Access Control

- Unauthorized users take advantage of poor access control to get around weak or absent authentication or authorization methods, such as identity and access management.
- For example, malicious actors take advantage of weak passwords to guess credentials.
- Strong access controls implement additional requirements, such as minimum password length, mixing upper and lower cases, the inclusion of punctuation or symbols, and frequent password changes

Vulnerability and Attacks (3)

Insider threats

- Insider threats are vulnerabilities attached to individuals or entities that already have some degree of access to and knowledge of an enterprise's IT environment.
- Insiders could include current and former employees, third-party vendors, and partners.
- Insider threats could be due to accidental errors, negligence, or malicious intent. Insider attacks via phishing and other social engineering techniques are common because humans remain the weakest link in a company's cybersecurity posture.

Vulnerability and Attacks (4)

Unsecured APIs

- Cloud APIs are the connective tissue that facilitates communication and data exchange between cloud software and applications, and API vulnerabilities are a prominent attack vector for threat actors.
- Examples of vulnerabilities associated with unsecured APIs include suboptimal access controls, weak authentication protocols, wrong rate limits, and accidental data exposure.

Vulnerability and Attacks (5)

Shadow IT

- Shadow IT is the use of user cloud assets **without the approval or support of user IT department**. There are several risks associated with this, including the financial impact of staff creating cloud workloads for personal use, data loss via unauthorized file-sharing services, and the use of unauthorized messaging services for communications.

Improper Encryption

- **Not followed rules in performing encryption and decryption**

Hypervisor Security (1)

Hypervisor Security

- Hypervisor security refers to the measures taken to protect the hypervisor and the VMs it manages from potential security threats.

Below are some key aspects of hypervisor security:

- **Access Control:** The hypervisor and its configuration parameters should only be accessible to authorized users. Through systems for authentication and authorization, access should be limited.
- **Isolation of Virtual Machines:** The hypervisor should make sure that VMs are kept apart from one another and that their operations don't compromise the security or stability of other VMs, the hypervisor, or both.

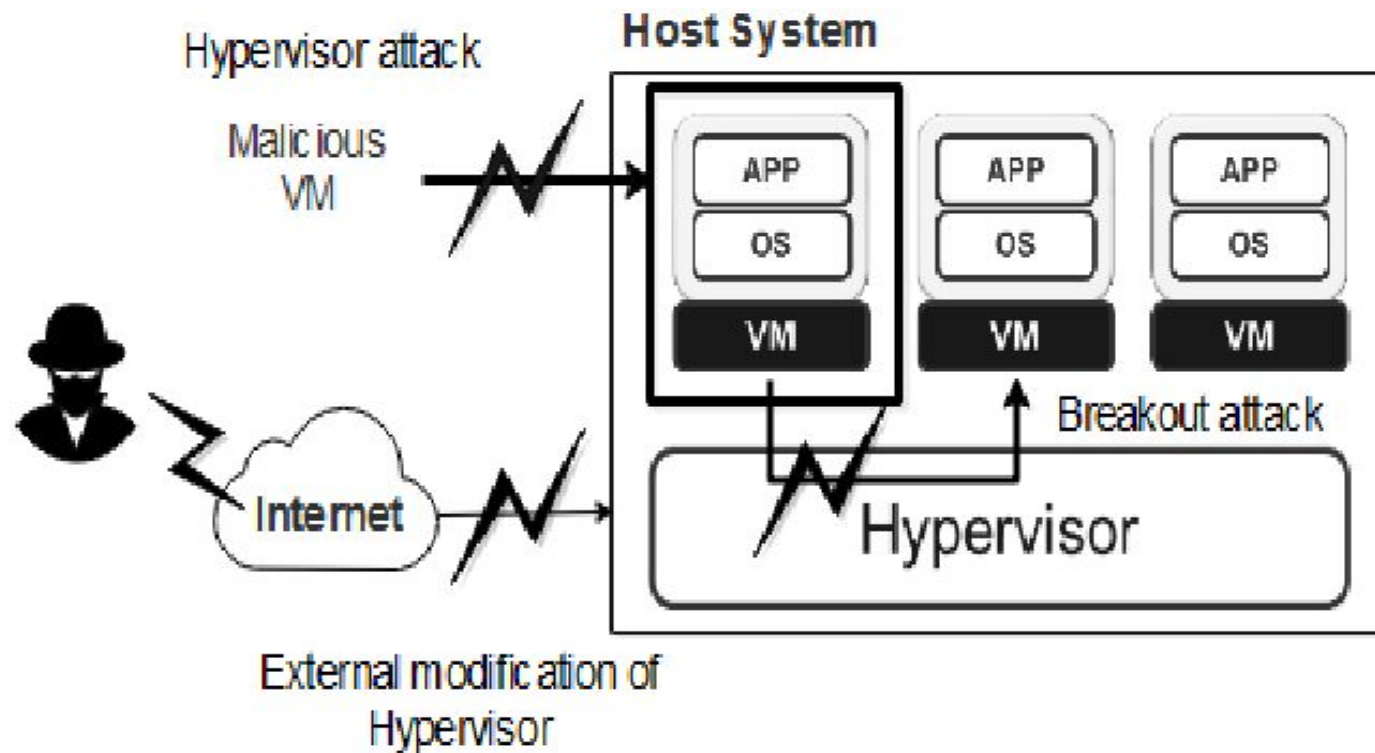
Hypervisor Security (2)

- **Threat Detection and Mitigation:** The hypervisor must include security tools that can quickly identify and address possible threats. For instance, firewalls, antivirus programs, and intrusion detection systems.
- **Patch Management:** To keep the hypervisor secure and stop vulnerabilities from being exploited, regular software upgrades and patches are essential.
- **Data Encryption:** To prevent theft and illegal access, sensitive data stored within VMs should be encrypted.

Hypervisor Security Against Attacks (1)

- **Limit users:** Limit the number of users on a local system.
- **Reduce attack surface:** Reduce the attack surface by removing unneeded functionality from the hypervisor.
- **Keep systems updated:** Keep the hypervisor software up to date.
- **Use a hypervisor security boundary:** A hypervisor security boundary can protect against attacks like denial-of-service and elevation of privilege. It also segments tenants for network traffic, storage, and more.
- **Use a reliable hypervisor:** Some hypervisors, like Xen Project, are considered safer and more reliable for security-first environments

Hypervisor Security Against Attacks (2)



Hypervisor Security Against Vulnerabilities (1)

Update and Patch

- Keep hypervisors, VMs, and related software up to date with the latest security patches and features.
- User can automate updates to reduce the risk of human error

Use firewalls and network segmentation

- Set up firewalls and network segmentations to protect hypervisors from public access.

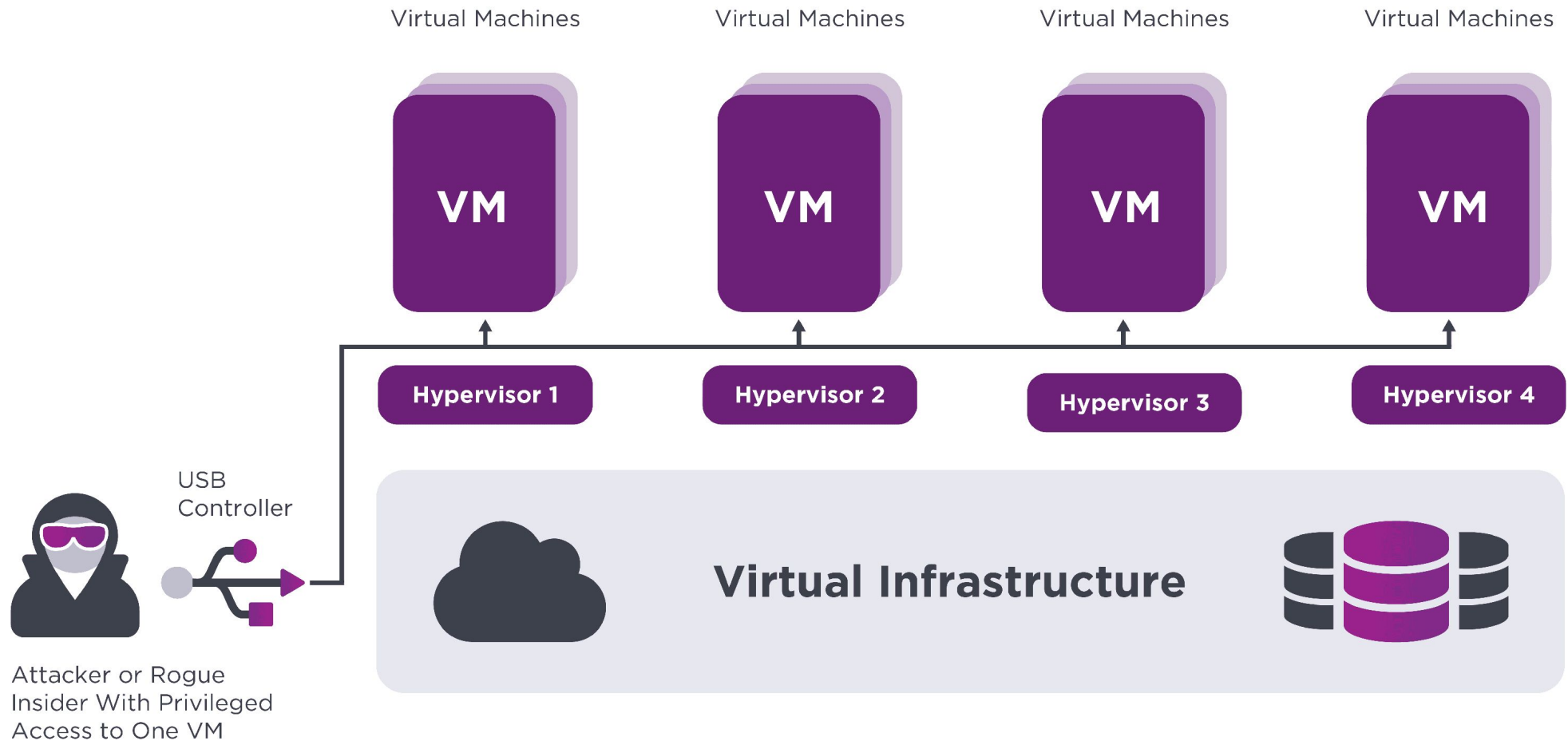
Create separate networks

- Keep the VM network separate from the management network.

Use intrusion prevention systems

- Install cutting-edge intrusion prevention systems to protect against network intrusions.

Hypervisor Security Against Vulnerabilities (2)



Hypervisor Security In Storage

- **Data Encryption:** Ensure that data at rest and in transit is encrypted. Use strong encryption standards to protect sensitive data stored in cloud environments.
- **Access Control:** Implement strict access controls using role-based access control (RBAC) to limit who can access or modify data.
- **Isolation:** Utilize separate storage accounts or containers for different tenants to prevent data leakage.
- **Monitoring and Logging:** Enable monitoring of storage access and usage to detect and respond to unauthorized access attempts.

Hypervisor Security In Hardware

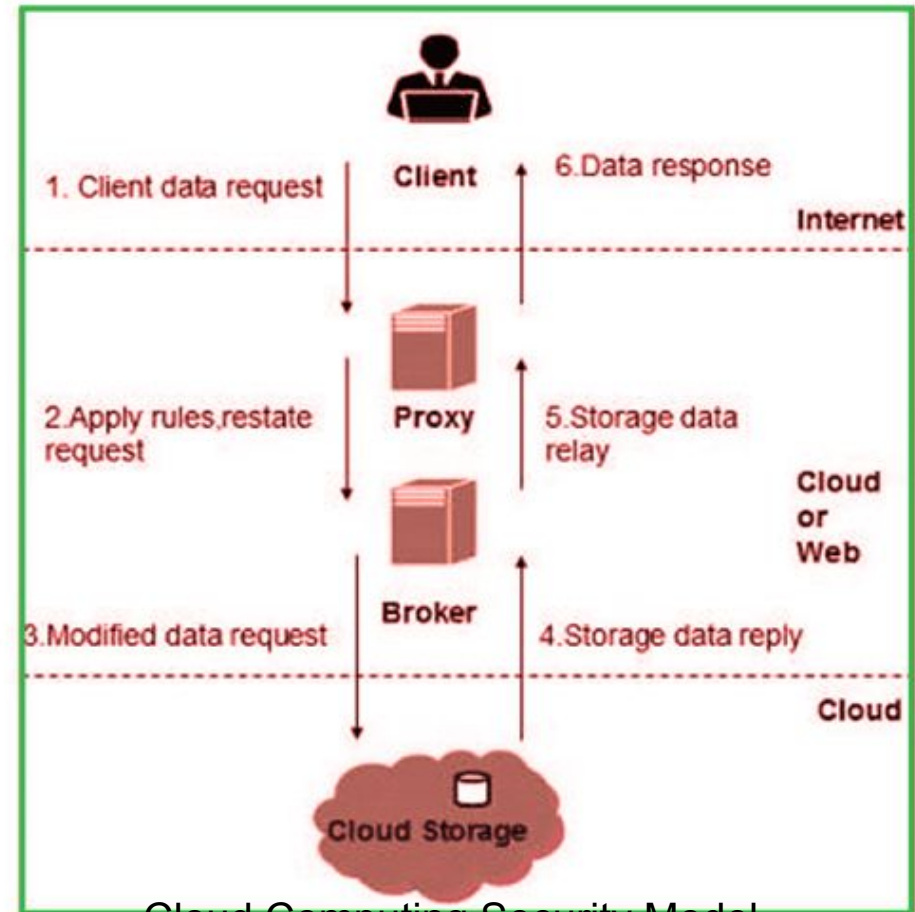
- **Secure Boot:** Ensure that the hypervisor and guest operating systems are loaded only from trusted sources.
- **Hardware-Based Security Features:** Utilize features like Intel TXT or AMD SEV to secure virtual machines (VMs) and ensure their integrity.
- **Physical Security:** Protect the physical servers hosting the hypervisors with access controls, surveillance, and environmental controls to prevent tampering.
- **Regular Patching:** Keep firmware and hardware components updated to protect against vulnerabilities.

Hypervisor Security In Networking

- **Segmentation:** Use network segmentation to isolate different workloads and minimize the attack surface.
- **Firewalls and Security Groups:** Deploy firewalls to control incoming and outgoing traffic and establish security groups that enforce policies based on the principle of least privilege.
- **Intrusion Detection and Prevention Systems (IDPS):** Implement IDPS to monitor network traffic for malicious activity and respond accordingly.
- **Secure APIs:** Ensure that APIs used for managing cloud resources are secure, authenticated, and have appropriate access controls in place.

Cloud Computing Security Mechanism

- Data Security
- Encryption
- Digital Signature
- Hashing



Cloud Computing Security Model

Cloud Computing Security Mechanism – Data Security (1)

- A broker has full access to the storage but does not have access to the client.
- A proxy does not have access to storage but has access to both the client and the broker.
- Example Working on a Brocade cloud storage access system
- When the client issues a request to access data:
- The client data request goes to the external service interface of the proxy.
- The proxy forwards the request to the broker.
- The broker requests the data from the cloud storage system.
- The cloud storage system returns the data to the broker.
- The broker returns the data to the proxy.
- Finally, the proxy sends the data to the client.

Cloud Computing Security Mechanism – Data Security Features (2)

Encryption

- Encrypts data at rest or in transit to make it unreadable without a key. This helps establish trust and anonymity, and is required by many privacy regulations.

Identity and access management (IAM)

- Helps prevent unauthorized access to cloud resources by controlling who can access them. IAM can be strengthened using multi-factor authentication (MFA), single sign-on (SSO), role-based access control (RBAC), and least privilege principles

Public key infrastructure (PKI)

- A framework that uses digital certificates to manage secure information exchange. PKI solutions can verify that data remains uncompromised and confidential

Back up data

- Organizations can back up their cloud data on the cloud itself.

Cloud Computing Security Mechanism – Encryption (3)

- Cloud encryption is a data security mechanism that protects cloud data by converting plaintext into unreadable ciphertext before storing it in the cloud. It's considered one of the most effective ways to protect data privacy and prevent cyberattacks.

Here are some benefits of cloud encryption:

Protects data

- Cloud encryption protects data from unauthorized access, even if it's lost or stolen.

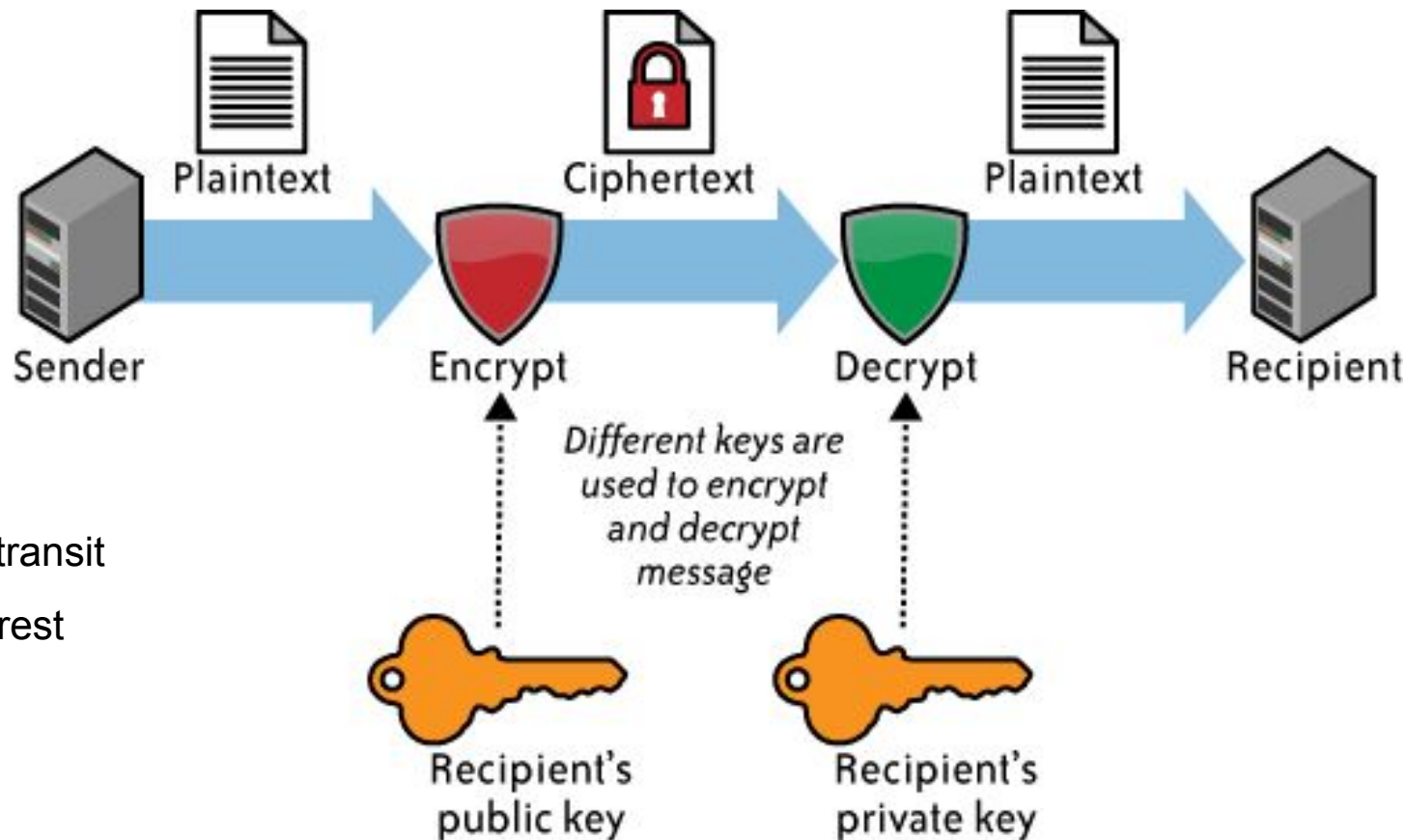
Complies with regulations

- Cloud encryption helps organizations comply with data privacy and protection regulations.

Prevents breaches

- Cloud encryption can help organizations avoid disclosing breaches or other security events.

Cloud Computing Security Mechanism – Encryption (3)



- Data-in-transit
- Data-at-rest

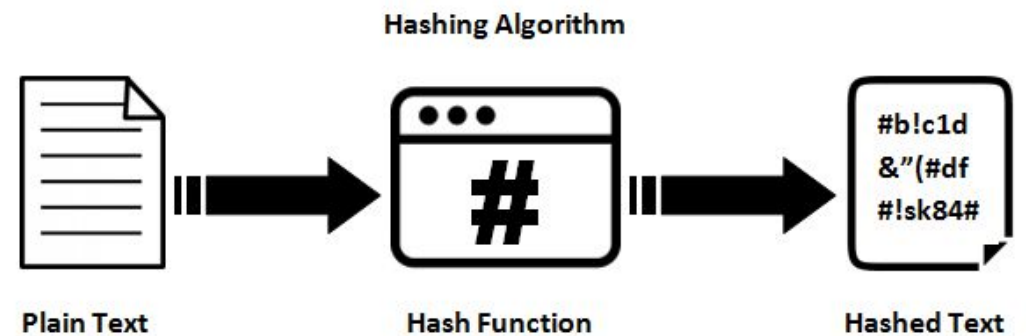
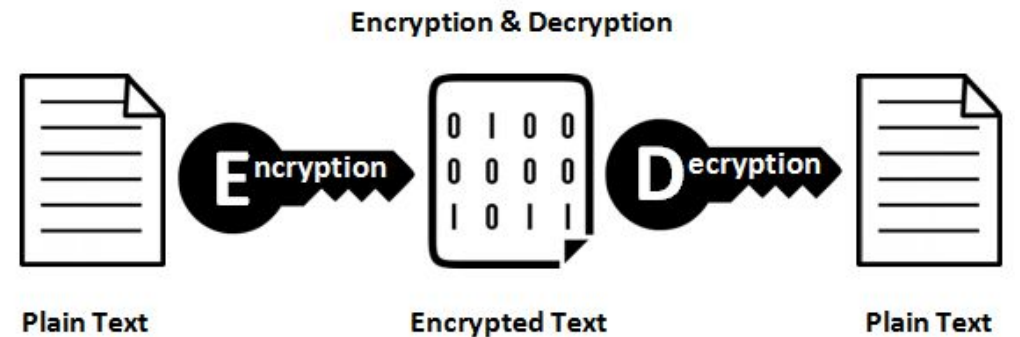
Asymmetric encryption

Cloud Computing Security Mechanism – Hashing (4)

- Hashing is a data security technique that uses a mathematical function to convert data into a unique, irreversible string of characters called a hash value.
- Hashing is a key security mechanism in cloud computing because it helps to protect sensitive data like passwords, messages, and documents.
- **Data integrity:** Hashing ensures that data is not corrupted or tampered with.
- **Security:** Hashing makes data unreadable and difficult to decrypt, even with advanced technology.
- **Validation:** Hashing can be used to validate the original data

Cloud Computing Security Mechanism – Hashing (4)

How Hashing Works



Cloud Computing Security Mechanism – Digital Signature (5)

- A digital signature is a cloud computing security mechanism that verifies the authenticity and integrity of data through cryptography.

How it works

A digital signature is created by applying a hashing algorithm to a message or document, and then encrypting the resulting hash with the sender's private key. The recipient uses the sender's public key to decrypt the signature and verify that the message is unaltered.

Benefits

Digital signatures provide several benefits, including:

Authenticity: Verifies that the sender is who they claim to be

Integrity: Ensures that the message or document has not been tampered with

Non-repudiation: Provides proof that the sender cannot deny sending the message or document

Cloud Computing Security Mechanism – Digital Signature (5)

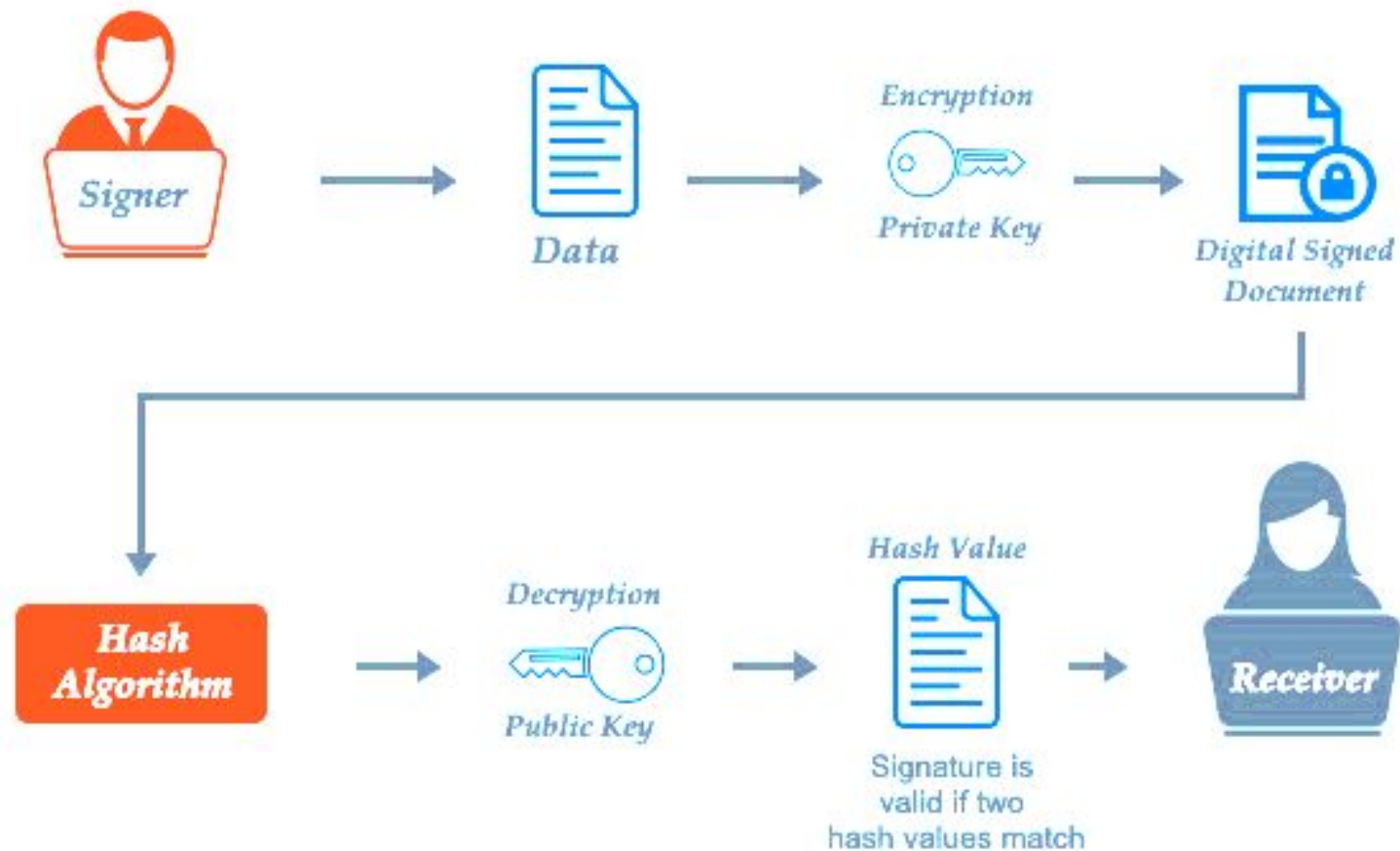
The following describes the flow for creating and validating a signature. The two participants in this workflow consist of the signer of data, and the data recipient.

1. The signer creates an asymmetric key that supports digital signing.

The signer can use this key to create multiple signatures.

2. The signer performs a private key operation over the data to create a digital signature.
3. The signer provides the data and the digital signature to the data recipient.
4. The recipient uses the public key portion of the signer's public/private key pair to verify the digital signature. If verification is unsuccessful, then the data has been altered.

Cloud Computing Security Mechanism – Digital Signature (5)



Security Features in Cloud Computing (1)

- **Centralized security:** Centralized security results in centralizing protection. As managing all the devices and endpoints is not an easy task cloud security helps in doing so. This results in enhancing traffic analysis and web filtering which means less policy and software updates.
- **Reduced costs:** Investing in cloud computing and cloud security results in less expenditure in hardware and also less manpower in administration
- **Reduced Administration:** It makes it easier to administer the organization and does not have manual security configuration and constant security updates.
- **Reliability:** These are very reliable and the cloud can be accessed from anywhere with any device with proper authorization.

Security Features in Cloud Computing (2)

TOP 7 SECURITY FEATURES OF MICROSOFT CLOUD

	Asset Protection	• Replicate data across multiple regions for high availability & disaster recovery. • Encrypt data at rest & in transit to protect it from unauthorized access.
	Application Protection	• Use Firewall & Azure Application Gateway to protect against web attacks. • Implement App Protection Policies to control user interactions.
	Access Protection	• Enforce Multi-factor Authentication for enhanced security. • Control access based on user roles. • Track user activity to investigate security incidents.
	Network Security	• Use Firewall to restrict traffic to trusted sources. • Isolate business resources using Virtual Network (VNet).
	Threat Protection	• Utilize machine learning for threat identification & mitigation. • Stay updated with threat intelligence data to block malware, phishing attacks,... • Regularly scan resources for vulnerabilities & fix them.
	Data Encryption	• Safeguard encryption keys in Azure Key Vault. • Encrypt Azure disks to secure data even when the disk is stolen. • Secure network traffic using TLS/SSL encryption.
	Antivirus	• Implement Advanced Threat Protection for malware detection & real-time threat alerts.

Virtualization Security

- Virtualization is the process of running multiple virtual instances of a device on a single physical hardware resource. Security virtualization is the process, procedure and policy that ensures that the virtualized hardware infrastructure is secure and protected.
- A virtualization system consists of a host operating system, a hypervisor, and a guest operating system.
- The host is the underlying hardware of the virtualization system that provides computing resources (physical interface cards, CPUs, memory, management ports, operating system, third-party software, hypervisor).
- All this enables the host to contain one or more Virtual Machines (or partitions) and share physical resources with them.

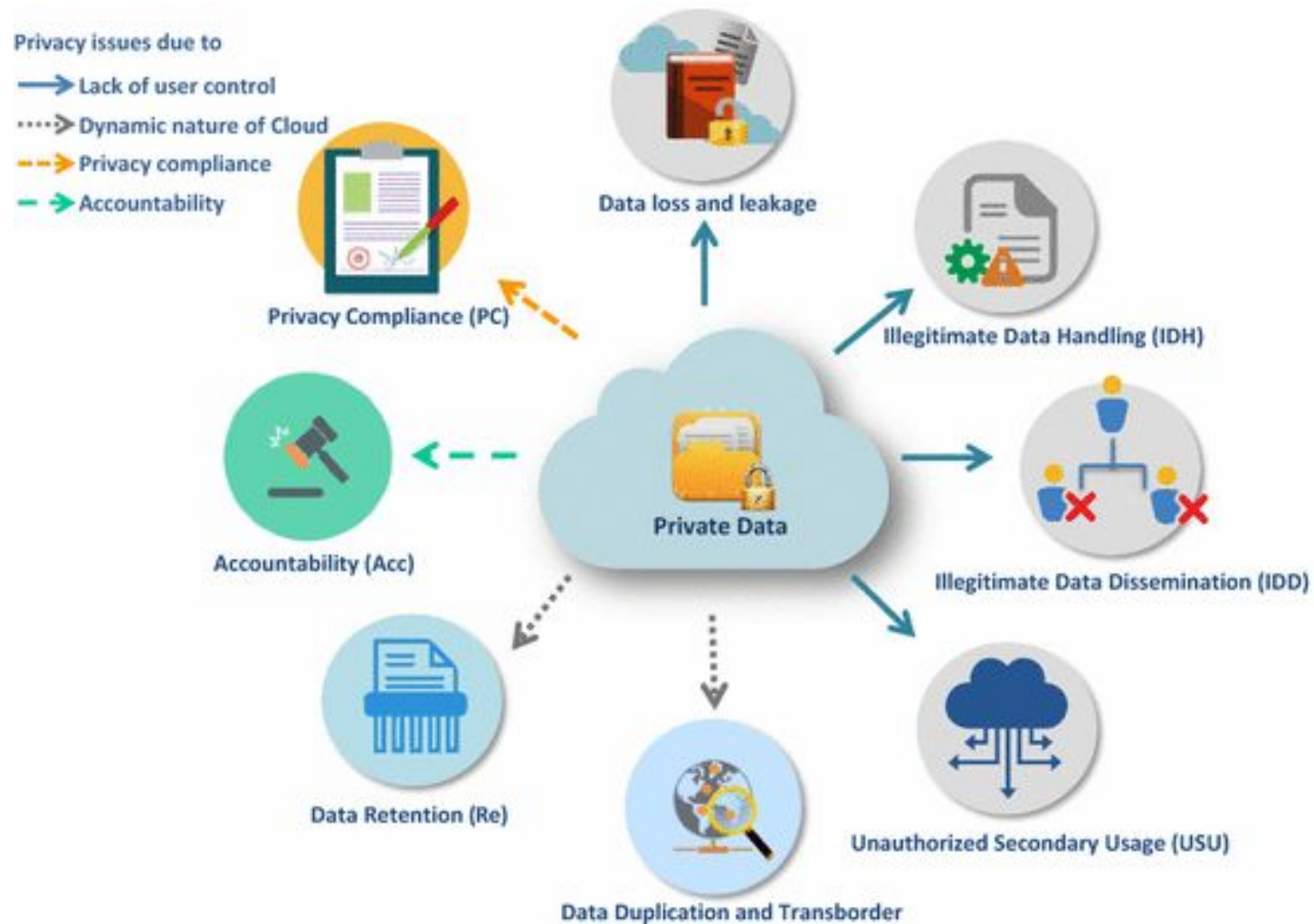
Privacy goals in Cloud Storage Services (1)

- Privacy is a fundamental human right that comprises the right to be left alone and demands the appropriate use and protection of personal information.
- Privacy protection in cloud computing environment is less of a technical issue and more of a policy and legal issue
- Policies are required to be frame to conform to the legal framework protecting the privacy of individual and organizations

Note

- The implementation of cloud computing paradigms violates privacy in different ways, such as misappropriation of confidential information
- Uncontrollable use of cloud services, data propagation, potential unauthorized secondary usage, trans-border flow of data and dynamic provisioning. Other privacy concerns are data retention regulation, outsourced data deletion, and privacy awareness breaches.

Privacy issues in Cloud Storage Services (2)



Security in Cloud Storage Services (1)

- Cloud data security is the practice of protecting data and other digital information assets from security threats, human error, and insider threats.

Goals or Best Practices



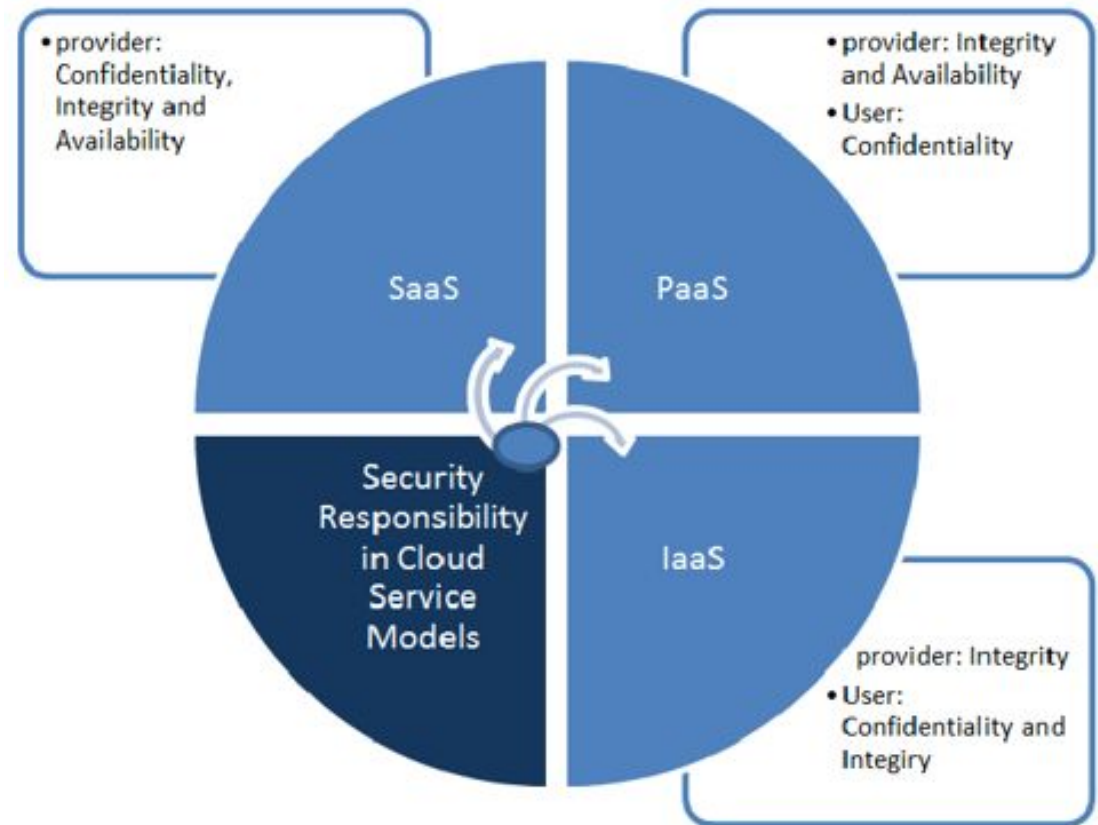
Cloud Data Protection Model

- Cloud data protection is the practice of securing a company's data in a cloud environment, wherever that data is located, whether it's at rest or in motion, and whether it's managed internally by the company or externally by a third party. This model has the following phases



Data Security Model

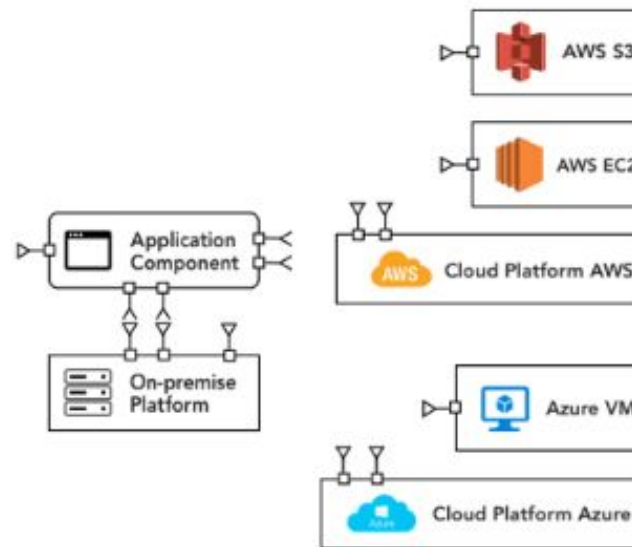
- Secure applications and data across multiple environments while maintaining complete visibility into all user, folder and file activity.
- Proactively identify and mitigate risks, such as security threats, suspicious user behavior, malware and others.



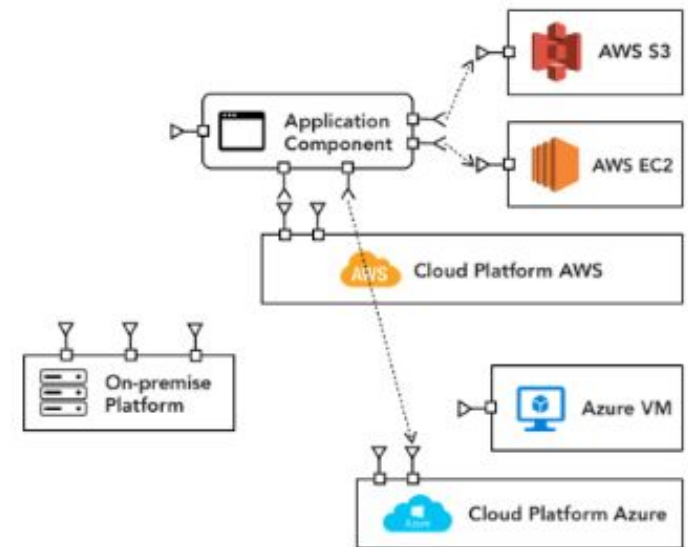
Multi Cloud Security

- Multi-cloud security refers to the policies, procedures, and technologies implemented to protect data, applications, and infrastructure across multiple cloud service providers.
- It ensures the confidentiality, integrity, and availability of resources in multi-cloud environments while maintaining compliance with relevant regulatory requirements.

Before Migration



After Migration



Security Properties in Multi Cloud (1)

Data Protection

- **Encryption:** Ensure that data is encrypted at rest and in transit across all cloud platforms. Use strong encryption algorithms and manage keys securely.
- **Data Segmentation:** Isolate data based on sensitivity and compliance requirements. Use different storage solutions or containers to minimize exposure.

Access Control

- **Identity and Access Management (IAM):** Implement robust IAM policies across all cloud providers. Use role-based access control (RBAC) and the principle of least privilege.
- **Single Sign-On (SSO):** Consider using SSO solutions to streamline authentication across multiple cloud services while enhancing security.

Security Properties in Multi Cloud (2)

Network Security

- Firewalls and Security Groups: Deploy firewalls to monitor and control traffic between clouds.
Define security groups to manage access based on workload requirements.

Virtual Private Network (VPN):

- Use VPNs to secure communication between on-premises infrastructure and multiple cloud providers.

Privacy Properties in Multi Cloud (1)

Data Confidentiality

- **End-to-End Encryption:** Ensure data is encrypted both at rest and in transit across all cloud services. Only authorized users should have access to decryption keys.
- **Zero-Knowledge Solutions:** Use services that provide zero-knowledge encryption, ensuring that even the cloud provider cannot access user data.

Data Segmentation and Isolation

- **Logical Separation:** Store sensitive data separately from less sensitive information. Use different accounts, containers, or services to isolate data.
- **Environment Segmentation:** Differentiate environments (e.g., production, development, testing) to reduce the risk of unauthorized access.

Privacy Properties in Multi Cloud (2)

Data Backup and Recovery

- **Regular Backups:** Ensure that backups are encrypted and stored securely across different cloud environments to protect against data loss.

Disaster Recovery Planning

- Develop a multi-cloud disaster recovery strategy to safeguard sensitive data and ensure privacy during recovery processes.

Steps for Enhancing Security in Cloud

- Choose a trusted cloud provider
- Use strong authentication
- Encrypt user data
- Keep user software up to date
- Monitor user cloud environment
- Use access controls
- Implement network security
- Train user employees

References

- <https://www.wiz.io/academy/common-cloud-vulnerabilities>
- <https://www.geeksforgeeks.org/hypervisor-security-in-cloud-computing/>
- <https://www.entrust.com/blog/2024/05/vmware-hypervisor-security-critical-vulnerabilities-related-to-usb-controllerStorage Security>
- <https://www.slideshare.net/slideshow/cloud-security-mechanisms/40066444>
- <https://www.oreilly.com/library/view/cloud-security-and/9780596806453/ch04.html>
- <https://cloud.google.com/kms/docs/digital-signatures>
- <https://www.juniper.net/us/en/research-topics/what-is-security-virtualization.html>
- <https://encyclopedia.pub/entry/17992>
- <https://stl.tech/blog/cloud-security-solution/>
- <https://www.fortinet.com/resources/cyberglossary/cloud-data-protection>
- [DOI:10.7763/IJET.2012.V4.375](https://doi.org/10.7763/IJET.2012.V4.375)
- <https://www.simform.com/blog/multi-cloud-architecture/>





THANK YOU